

Daily Algebra Drill #5

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Section	Topic	Questions	Target time
A	Rapid Recognition	10	2 min 30 sec
B	Manipulation Drills	10	8 min
C	Substitution & Structure	8	10 min
D	Challenge	5	15 min

New toolkit today: Partial fractions with quadratic denominators · Elementary functional equations · Cauchy–Schwarz (Engel form) · Conjugate surd pairs revisited.

Attempt each question before checking answers. Time yourself. No calculators.

Section A Rapid Recognition

(≤15 s each)

No expansion. Recognise the pattern, write the answer.

1. Simplify: $\frac{x^2 - 9}{x^2 + 5x + 6}$.
2. Rationalise: $\frac{4}{\sqrt{7} - \sqrt{3}}$.
3. Given $x + y = 3$ and $xy = -4$, find $(x - y)^2$.
4. Factorise: $x^3 - x^2 - 4x + 4$.
5. Without expanding, find the coefficient of x^2 in $(1 + x)^4$.
6. Given $\alpha + \beta = -3$ and $\alpha\beta = -10$, write down the quadratic with roots α and β .
7. Simplify: $\sqrt{50} - \sqrt{18} + \sqrt{8}$.
8. Given $a + b = 7$ and $a^2 + b^2 = 29$, find $a^3 + b^3$.
9. Solve: $|2x - 3| = 7$.
10. For $x > 0$, find the minimum of $\frac{(x + 4)^2}{x}$.

Section B Manipulation Drills

(≤50 s each)

Fractions, structure, identities. Minimum working.

1. Decompose into partial fractions: $\frac{5x + 1}{(x + 1)(x - 2)}$.
2. Decompose into partial fractions: $\frac{3}{x(x^2 + 3)}$.

3. Simplify: $\frac{1}{\sqrt{n} + \sqrt{n+1}} + \frac{1}{\sqrt{n+1} + \sqrt{n+2}}$ and hence evaluate $\sum_{k=0}^8 \frac{1}{\sqrt{k} + \sqrt{k+1}}$.
4. Factorise: $a^4 + a^2 + 1$ over $\mathbb{Z}$. (*Hint: add and subtract a^2 .*)
5. Given $a + b = s$ and $ab = p$, express $(a - b)^4$ in terms of s and p .
6. Simplify: $\frac{x^2 - 2x + 1}{x^2 + 2x - 3} \div \frac{x^2 - 1}{x^2 + 4x + 3}$.
7. The roots of $x^2 + px + q = 0$ are r and s . Find in terms of p and q the quadratic with roots r^2 and s^2 .
8. Solve the simultaneous equations: $\frac{1}{x} + \frac{1}{y} = \frac{5}{6}$ and $\frac{1}{x} - \frac{1}{y} = \frac{1}{6}$.
9. Prove: $\frac{1}{1 \cdot 3} + \frac{1}{3 \cdot 5} + \frac{1}{5 \cdot 7} + \cdots + \frac{1}{(2n-1)(2n+1)} = \frac{n}{2n+1}$.
10. If $f(x) = \frac{ax+b}{cx+d}$ and $f(f(x)) = x$ for all x , what condition must a, b, c, d satisfy?

Section C Substitution & Structure

(≤90 s each)

One clean substitution is worth ten lines of algebra.

1. Solve: $\left(x^2 + \frac{1}{x^2}\right) - 7\left(x - \frac{1}{x}\right) - 10 = 0$.
2. For positive reals a, b, c with $a + b + c = 3$, prove $\frac{a^2}{b} + \frac{b^2}{c} + \frac{c^2}{a} \geq 3$ using the Cauchy–Schwarz (Engel/Titu) form: $\frac{x^2}{p} + \frac{y^2}{q} + \frac{z^2}{r} \geq \frac{(x+y+z)^2}{p+q+r}$.
3. Solve the system: $x^2 + xy = 12$, $xy + y^2 = 6$.
4. Find the range of values of k for which $kx^2 - 4x + k - 3 = 0$ has two distinct real roots.
5. Simplify $\sqrt{3+2\sqrt{2}} + \sqrt{3-2\sqrt{2}}$ without a calculator.
6. If $x - \frac{1}{x} = 4$, find the exact value of $\sqrt{x} - \frac{1}{\sqrt{x}}$.
7. Given $a, b > 0$ with $\frac{1}{a} + \frac{1}{b} = 1$, show that $(a-1)(b-1) = 1$ and find the minimum of $a+b$.
8. Solve: $(x^2 - x - 1)^2 = (2x + 1)^2$.

Section D Challenge

(SMC / BMO1 difficulty)

No brute force. Each question rewards one key observation.

1. A function f satisfies $f(x) + f\left(\frac{1}{1-x}\right) = x$ for all $x \neq 0, 1$. Find $f(x)$. (Apply the substitution twice more to get three equations.)

2. For positive reals, prove the Cauchy–Schwarz inequality in **Engel (Titu) form**:

$$\frac{a^2}{x} + \frac{b^2}{y} + \frac{c^2}{z} \geq \frac{(a+b+c)^2}{x+y+z}.$$

Hence find the minimum of $\frac{1}{x} + \frac{4}{1-x}$ for $0 < x < 1$.

3. Find all functions $f : \mathbb{R} \rightarrow \mathbb{R}$ satisfying $f(x+y) = f(x) + f(y)$ and $f(1) = 3$. (First find f on $\mathbb{Z}$, then state any assumptions needed for $\mathbb{R}$.)
4. Show that among any 5 integers, there exist at least 2 whose difference is divisible by 4. Hence prove that for any 5 integers, two of them satisfy $a^2 \equiv b^2 \pmod{4}$.
5. Solve for integers: $x^2 - y^2 = 2024$.

“A formula you have derived yourself is worth a hundred you have memorised.”

Found a mistake or want to contribute a sheet?

Visit the open-source project at github.com/The-CerealDev/speed-maths